

Replication Instructions

Step 1: Environment Setup

Verify Python 3.8 or higher is installed, then install all dependencies:

```
python --version # must be 3.8+(I used 3.14.2)
pip install numpy pandas scikit-learn scipy matplotlib
```

Step 2: Download the Datasets

Download the 5 CSV files from the official lab repository or my github repository:

<https://github.com/ideas-labo/ISE/tree/main/lab1> or
<https://github.com/aasfimanwar/ise-coursework> (data folder)

Note: I have changed the file name from incubator-mxnet.csv to mxnet.csv in my repository

Files needed (Note: MXNet file must be named mxnet.csv):

- TensorFlow.csv
- PyTorch.csv
- Keras.csv
- Mxnet.csv
- Caffe.csv

Place all five files in a folder named data/.

Step 3: Download the Code

Download classify.py from the repository linked in the Artifact section of the report. Your directory should look like this:

```
project/
  classify.py
  data/
    TensorFlow.csv
    PyTorch.csv
    Keras.csv
    Mxnet.csv
    Caffe.csv
  results/    <- created automatically
```

Step 4: Run the Experiment

```
python classify.py --data_dir ./data --out_dir ./results
```

This will:

- Run 30 independent train/test splits per project (70/30 stratified split)
- Evaluate both Baseline (NB+TF-IDF) and Proposed (SVM+TF-IDF)
- Perform Wilcoxon signed-rank tests and compute Vargha-Delaney A
- Save all figures, the summary CSV, and raw JSON scores to ./results/

Expected runtime: 3-5 minutes on a standard laptop.

Step 5: Verify Results

Open `results/results_summary.csv`. The values in the paper correspond to the mean column for each project and metric. Minor floating-point differences of ± 0.001 are acceptable due to platform, version or any such differences.

The raw per-run scores for all 30 repetitions are in `results/raw_results.json` for full verification.

Reproducibility Notes

- All random seeds are fixed seed equals the repeat index (0 through 29).
- The stratified split ensures positive samples appear in both train and test sets.
- No external data sources or pre-trained models are used.
- Results are fully deterministic given the same dataset and Python/scikit-learn versions.